

# Axis-Parallel Rectangles in Dense Sets

GREEN-051

L1 - TRACTABLE

OPEN

Geometry

## ORIGINAL PROBLEM STATEMENT

Suppose  $A$  is an open subset of  $[0, 1]^2$  with measure  $\alpha$ . Are there four points in  $A$  determining an axis-parallel rectangle with area  $\geq c\alpha^2$ ?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>  
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>